

Simple Reaction Task and Stroop Effect
PSY 310: Lab in Psychology
Lab Report

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Introduction

Reaction time is the lapse between perceiving a stimulus and responding to through detection, processing and response. It is a measure of brain processing, cognitive health and mental agility. The study of reaction time is called mental chronometry which helps break down the invisible steps behind reactions to stimulus. A simple reaction time task (SRT) measures speed of response to a single stimulus. In this experiment, participants see one stimulus, make a predetermined response (only one correct response). The timings are varied with pauses to create unpredictability and get raw reaction times. The results of a simple reaction time task are influenced by age, arousal, fatigue, complexity of task, practice and individual differences.

In addition to SRT, cognitive processing can also be tested when competing information is presented as seen in Stroop Effect. John Ridley Stroop (1935) demonstrated through his work how difficult it is to overcome an automatic cognitive process like reading. The Stroop Task involves asking participants to read the ink of the color written rather than the word, for example the word 'blue' written in purple ink. As stroop demonstrated, it took participants significantly longer to name the ink color due to interference effects. Further research by George Klein made the Stroop task important for studying cognitive control - the ability to direct thinking and behaving to achieve a goal.

Both SRT and Stroop Task are different approaches to examining response performance and cognitive processing. While SRT primarily examines speed of response to single stimulus, the Stroop Task involves conflicting information providing insight into effects of cognitive interference on response performance. Comparing performances across these tasks can help demonstrate how the presence of competing information can influence accuracy and reaction time. Performing both experiments can provide an understanding of basic response speed and cognitive processing when automatic and controlled responses compete.

Method

Participant

The study had one subject, who was a female, 43 years old. Their informed consent was taken prior to the experiment and confidentiality was assured.

Materials

The study was conducted using PsychoPy v2026.2.3. It was conducted on a laptop with a monitor of dimensions 12.5", with a resolution of 1920x1080 pixels and a refresh rate of 60Hz.

Procedure

In this Simple Reaction Time Task the participants were presented with an instruction routine regarding the task and required response. The participants were told to respond by pressing the “Y” key when the target stimulus, white square, appeared on the screen. Following the instructions, a trial routine was presented. A fixation symbol, white star, was presented initially followed by the appearance of a white square at 0.5 seconds. The response of the participant was recorded when they pressed the specified Y key. A loop was created around the trial routine with the loop set to sequential and the number of repetitions was set to 5.

The Stroop Task was used to examine response performance in the presence of cognitive interference. Participants were required to respond according to the color of the stimulus while disregarding potentially competing information presented by the word. A textual component was used to present the color-word stimuli. The written color word and color of word was varied around two conditions; incongruent and congruent. In the congruent condition the word RED was presented in red while in the incongruent condition it was presented in blue. The same was true for BLUE. The participant was required to press ‘b’ for blue inked text and ‘r’ for red inked text while ignoring the meaning of the word. The response accuracy was also recorded by comparing the participant’s response with the correct answer specified for each trial. A loop was added in sequential order to present the different color word combinations through an excel file which provides trial by trial information. The file contained parameters corresponding to the written word, color of stimulus, experimental condition and correct response.

The experiment was terminated after the participant had completed both tasks. The recorded responses were used for subsequent analysis of reaction time and accuracy. For SRT response speed was assessed using the recorded reaction times while for Stroop Task performance was examined by comparing accuracy and reaction time between the congruent and incongruent conditions.

Results

Simple Reaction Task

Trial	Reaction Time (ms)
1	349
2	559
3	2660

4	421
5	362
Mean	870.02

The Simple Reaction Time Task showed a mean reaction time of 870.02 ms across the 5 trials with 100% accuracy of response from the participant.

Stroop Task

Trial	Word	Color	Condition	Correct Response	Participant Response	Accuracy	Reaction Time (ms)
1	RED	red	congruent	red	red	Yes	487
2	RED	blue	incongruent	blue	red	No	756
3	BLUE	blue	congruent	blue	blue	Yes	521
4	RED	blue	incongruent	blue	blue	Yes	693
5	BLUE	red	incongruent	red	red	Yes	718
6	BLUE	blue	congruent	blue	blue	Yes	498
7	RED	red	congruent	red	red	Yes	536
8	BLUE	red	incongruent	red	red	Yes	642
9	BLUE	blue	congruent	blue	blue	Yes	512
10	RED	blue	incongruent	blue	red	No	681

Condition	No. of Trials	Mean RT	Accuracy
Congruent	5	510.8	100%
Incongruent	5	698	60%
Overall	10	604.4	80%

Stroop effect:

$$\begin{aligned} & \text{Incongruent Mean RT} - \text{Congruent Mean RT} \\ & 698 - 510.8 \\ & = 187.2 \end{aligned}$$

The mean reaction time for the congruent trials was 510.8ms whereas for the incongruent trials it was 698 ms resulting in a Stroop interference effect of 187.2ms. Accuracy was 100% for congruent trials and 60% for incongruent trials with overall accuracy of 80%. The participant therefore demonstrated slower and less accurate responses when the written word and stimulus color were incongruent.

Discussion

In the Simple Reaction Time Task the participant showed an average reaction of 382.6 ms which is the time taken to initiate a response following the presentation of target stimulus. As the reaction times were consistent across trials it can be concluded that the participants were able to respond to the stimulus with a relatively stable level of processing speed. In contrast, the mean reaction time in the Stroop task was 510.8 in congruent condition and 698.3 ms in incongruent condition showing a slower reaction time suggesting the presence of a Stroop interference effect where conflicting information required more cognitive processing and increased response time.

The increased reaction time in the incongruent condition can be due to a conflict between controlled and automatic processes, i.e, reading a word is an automatic process but identifying and naming the ink of the word requires greater attentional control, making it difficult to provide an appropriate response. The participant required more time to identify the relevant color and produce the accurate response due to competition in processing. The presence of interference is further strengthened by the lower accuracy observed in the incongruent condition.

The Simple Reaction Task was a straightforward single stimulus response experiment requiring the participant to respond to the presence of a target unlike the Stroop task that required selective attention to color of stimulus while inhibiting automatic response to respond to meaning leading to a longer reaction time in the incongruent Stroop trials. The longer response times reflect the additional attentional and inhibitory demands imposed by the Stroop task. The results show the importance of reaction time as an indicator of cognitive processing. Simple reaction Time Task provided an estimate of basic response speed while the Stroop Task provided insight into higher order processes like response inhibition, selective attention and influence of competing information on speed and accuracy of responding. Reaction time measures are useful in showcasing cognitive demands involved in different types of tasks.

These results have considerable limitations like the small sample size of one participant which limits generalizability of findings, individual difference in reaction time, familiarity with

keyboard responses, attention, fatigue effect, small number of trials which do not provide sufficiently reliable estimate of participant response times. There could have been practice or anticipation of stimulus response mapping so a larger sample and more trials would be better to establish more reliable conclusions regarding the effects of cognitive interference on reaction time.

References

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